

SmolVLM for Dense Video Captioning

Week 8: Internship Update

Code updates: MaxInfo

- Adjusting and optimizing
- Post-sampling included
- Pre-sampling included
- Shot-detection excluded from the MaxInfo algorithm

Code updates: MaxInfo

- Normalization after TruncatedSVD:

- Values quite big:

$$\text{vol}(A) = \sqrt{\det(AA^T)}$$

- StandardScaler: $N(0,1)$
- MinMaxScaler: $[0,1]$
- l2 normalization: 2-norm of the row = 1

Code updates: MaxInfo

- MaxVol algorithm:
 - Paper: <https://arxiv.org/pdf/1502.07838>
 - GitHub: <https://bitbucket.org/muxas/maxvolpy/src/master/>
 - Documentation:
https://pythonhosted.org/maxvolpy/generated/maxvolpy.maxvol.rect_maxvol_svd.html#maxvolpy.maxvol.rect_maxvol_svd
 - Both with and without included SVD
 - Tested without SVD (separately implemented)

$$A \approx CA[\text{pivots}]$$

Code updates: CSTA

- Optimizing
- Including presampling and postsampling
- Now works on all frames